

Question		Answer	Mark	Mark scheme	Additional guidance
14	(a)(i)	0.83	B1	for a tangent drawn at $t = 15$	Working: $7.5 \div 9 = 0.83....$ No tangent scores 0 marks
			M1	full method to use the tangent to find the gradient (eg $7.5 \div 9$)	This mark can be awarded if the tangent is drawn at $t \neq 15$ Working may be seen on the diagram
			A1	for answer in the range 0.6 to 1.0	
	(ii)	Statement	C1	statement Acceptable examples acceleration rate of change of speed increase in speed over time Not acceptable examples rate of change m/s/s increase in speed	
	(b)	220	P1	for splitting the area into strips and correct process to find the area of one strip, eg $\frac{5 \times 4}{2} (=10)$ or $\frac{(4+12)}{2} \times 5 (=40)$ or $\frac{(12+18)}{2} \times 5 (=75)$ or $\frac{(18+20)}{2} \times 5 (=95)$	Working 4, 12, 18, 20
			P1	for a complete process using at least 4 strips to find the area under the curve eg, “10” + “40” + “75” + “95”	Allow one error in the reading of speeds
			A1	for answer in the range 215 to 225 from correct working using at least 4 strips	